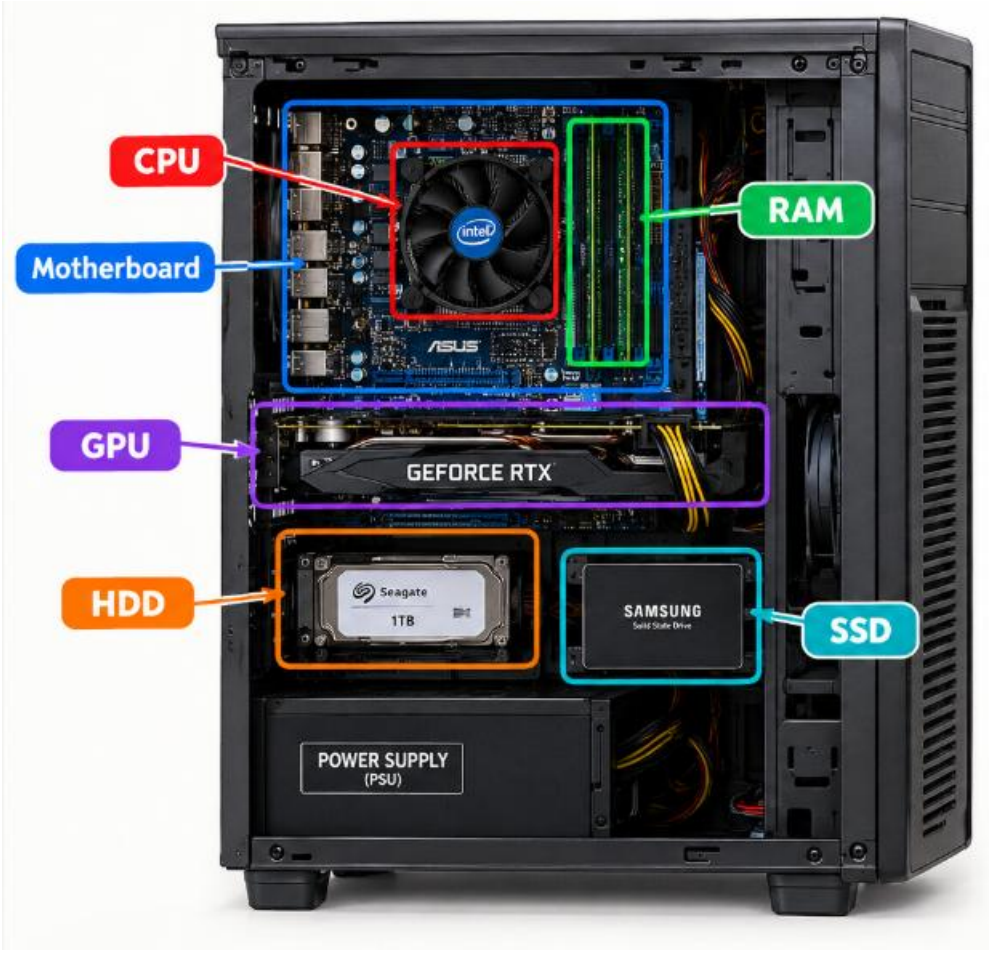
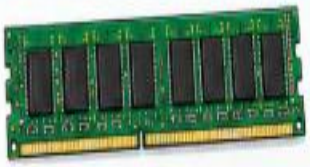
Overview Of Hardware Components

1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



Desktop PC - Main Components

S.No.	Part	Drawing	Use / Function
1.	Motherboard		• Main circuit board. • Connects all components (CPU, RAM, storage, GPU, etc). • Provides power and data communication.
2.	Processor (CPU)		• The brain of the computer. • Processes all instructions and data.
3.	RAM		• Temporary memory. • Stores data that CPU is currently using.
4.	SSD (Solid State Drive)		• Stores data permanently. • Faster than HDD. • No moving parts.
5.	HDD (Hard Disk Drive)		• Stores large amount of data permanently. • Slower than SSD. • Uses moving parts (disk & head).
6.	GPU (Graphics Card)		• Handles graphics and video processing. • Improves performance in games and graphics work.

2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower in reading and writing data.	Much faster in reading and writing data.
Durability	Less shock-resistant because it contains moving parts.	More shock-resistant because it has no moving parts.
Technology	Uses spinning magnetic disks to store data.	Uses flash memory to store data.
Boot Time	Takes more time to start the computer.	Starts the computer much faster.
Storage Capacity	Available in large capacities at a lower price.	Available in various capacities but usually costs more per GB.
Price	Generally cheaper than an SSD.	Generally more expensive than an HDD for the same storage capacity.
Power Consumption	Generally consumes more power.	Generally consumes less power.
Typical Use Cases	Used for storing large files, backups, movies, and documents.	Used in laptops, gaming PCs, and computers that require fast performance.
Noise	May produce noise due to moving parts.	Operates silently because it has no moving parts.

3. Write a short explanation describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

-> CPU (Central Processing Unit)

The CPU is the **brain of the computer**. When you play a game like PUBG or Free Fire, the CPU calculates player movements, controls enemy actions, processes game rules, and handles other instructions. For example, when you press a key to move your character or shoot an enemy, the CPU processes these actions and helps the game respond correctly.

-> RAM (Random Access Memory)

RAM is the computer's **temporary memory**. When you start a game, RAM stores the data the game needs quickly, such as game information, character data, and other frequently used resources. For example, when you move from one area of the game map to another, RAM helps keep the required data readily available so the game can run smoothly.

-> GPU (Graphics Processing Unit)

The GPU is responsible for **creating and displaying the game's visuals** on your screen. It processes graphics such as characters, buildings, shadows, lighting, explosions, and other visual effects. For example, when you play PUBG, the GPU renders the map and characters and displays the action on your screen. A powerful GPU can provide better visual quality and smoother gameplay with higher frame rates.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

If I want to upgrade my PC to run the latest FIFA game smoothly, I would consider upgrading the following hardware components:

-> CPU (Central Processing Unit):

I would upgrade to a faster CPU because it processes game instructions, player movements, and other calculations. A powerful CPU helps the game run smoothly.

-> GPU (Graphics Processing Unit):

I would upgrade to a powerful GPU because it handles the game's graphics, animations, lighting, and visual effects. A better GPU provides higher frame rates and better graphics quality.

-> RAM (Random Access Memory):

I would increase the RAM to ensure the game has enough memory to run smoothly. Sufficient RAM also helps reduce performance problems when other applications are running in the background.

-> SSD (Solid State Drive):

I would install an SSD to reduce game loading times and help the computer start faster. It also provides faster access to game files compared to an HDD.

-> Power Supply Unit (PSU):

If I install a more powerful CPU or GPU, I would consider upgrading the PSU to provide sufficient and stable power to the new components.